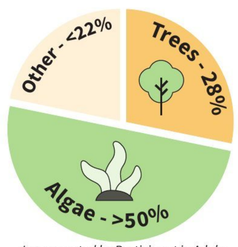
Continual Reinforcement Learning for Microalgae Bioreactor Control: Outperforming Static and PID Systems in Carbon Capture

Can AI Teach Itself to Grow Algae Faster?

Sia Gupta

Introduction

Global Oxygen Production[3]



Icons created by Participant in Adobe Illustrator

0
arable land

11x
biofuel production[2]

0
net CO_2 emissions

What's stopping current systems?



(Eric W, 2012, https://offgridworld.com/)

- **Static:** receive same nutrients and lights *every* day
- Only at **20%** of theoretical efficiency[1]
- Algae biomass production still costs more than crude oil[1]

How can we grow algae adaptively?

Continual RL

Reinforcement Learning (RL) is an AI approach that learns through trial and error. Standard RL learns from fixed training data. Continual RL *never* stops learning.



Research Question

How can an embedded reinforcement learning algorithm autonomously regulate environmental conditions in an algae bioreactor to maximize carbon capture efficiency by continuously learning?

Engineering Goals

Goal One: Simulation

Train a PPO agent in a Monod + Haldane simulation before hardware deployment.

- Random exploration would crash culture
- 200,000 steps at 30-min intervals = 11+ years on real hardware

Criteria: convergence within 200,000 steps

Goal Two: Continual Learning

Deploy trained agent onto live hardware with policy updating from real DO sensor data. Perform actions through LEDs for light intensity and servo controlled syringes for nutrient dosing. Run continuously for five day cultivation cycles.

Target Performance

Metric	Target
DO improvement vs. static	>10%
Mean biomass vs. static	>20%
DO improvement vs. PID	>5%
Mean biomass vs. PID	>10%
Simulation convergence	<200k steps

Variables

Independent Variable:
- Control Method (RL, PID, Static)

Dependent Variables:
- Dissolved Oxygen (mg/L), biomass (g)

Controlled Variables:
- Culture volume, species, nutrient medium, hardware, trial duration, dark period

Uncontrolled Variables:
- Room temperature, CO_2, humidity

Key Materials

- Optical DO Probe
- WS2812B LED Strip
- Micro-servo syringe
- Raspberry Pi 4B
- Chlorella vulgaris
- F/2 nutrient medium

Methods

Prepare Culture — C. vulgaris culture was inoculated into F/2 medium

Training Sim — PPO agent trained 200K steps using Monod + Haldane models

Setup Reactor — Identical hardware deployed side by side

Deploy Model — Deployed agent on Pi with continual learning

Collect Data — DO logged every 30 min, measured wet mass daily.

Culture Preparation

Component	Volume
C. vulgaris stock culture	200 ml
Distilled water	~49 ml
F/2 concentrate	1 ml
Total per reactor	250 ml



Photos taken by Participant

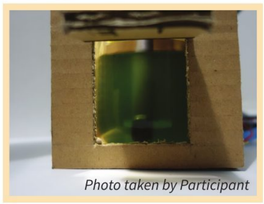


Photo taken by Participant



Photo taken by Participant

Iterations

Version	Sim Model	RL Algorithm	Notes
1	Droop	Augmented Random Search (ARS)	• Droop required unmeasurable internal cell quota • ARS couldn't handle two coupled continuous outputs
2	Andrews	ARS	• Andrews ignored light dynamics
3	Monod Kinetics & Haldane Light Inhibition	Proximal Policy Optimization (PPO)	• Biologically valid & stable $O_2 = P_{max} \cdot \frac{I}{I + K_{sL} + \frac{I^2}{K_i}} \cdot \frac{S}{K_{sN} + S}$

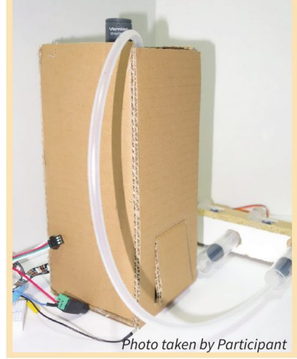


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Wiring Schematic

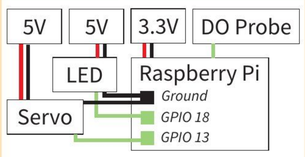


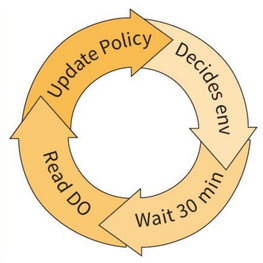
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Space

Observation: DO level, trend, average, nutrient budget, time
Action: LED Intensity, Nutrient Injection
Reward: (O_2 rate / 5.0) − (light × 0.005) + 10 if O_2 > 60

Static & PID

- **Static:** fixed at 75% light and 2ml / 12 hours
- **PID:** adjusts light and nutrients every 30 minutes based on fixed rules and error calculations



Results

+10.17% mean DO from static

1.96x biomass increase (static)

+4.3% mean DO from PID

1.42x biomass increase (PID)

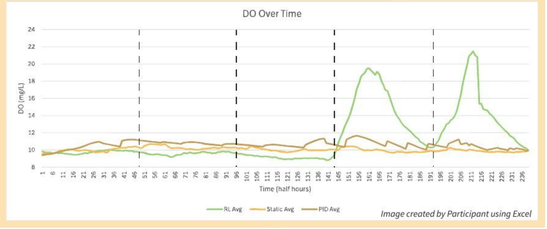


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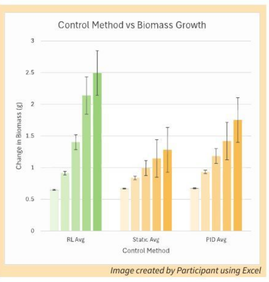


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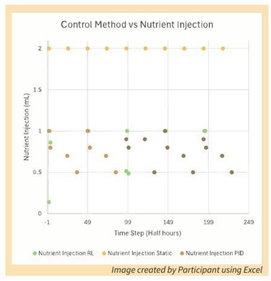


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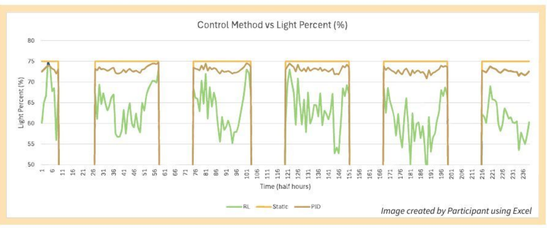


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Metric	RL	Static	PID
Mean DO	11.05	10.03	10.60
Peak DO	23.84	13.92	11.90
Biomass Inc	2.49	1.28	1.75

Data Analysis

Metric	RL v Static p	RL v PID p
Mean DO (mg/L)	0.045	0.055
Biomass (g)	0.006	0.002

A two-sample t-test was conducted on per-trial mean dissolved oxygen values and final biomass measurements to assess statistical significance with unequal variance configuration (Type 3).

Non-overlapping standard error bars on biomass confirm visual separation between all three conditions.

Days	RL Avg DO	Static Avg	PID Avg
1-2	9.68	10.13	10.60
3-4	11.38	9.99	10.70
5	13.65	9.88	10.364

Engineering Goals:
- Statistical improvement over static - Met (p values < 0.05)
- Statistical improvement over PID - Partial (met for biomass)
- Simulation Convergence - Met (converged with reward increasing from 1.86 -> 29.28)
- Continual learning: updated every 48 steps autonomously; RL agent improved over time (seen in daily averages and peaks in graph)

Observation:
- RL peak DO reached 23.84 mg/L compared to static peak at 13.92 mg/L, 71% higher maximum
- RL achieved higher mean DO using less average light intensity and lower nutrient dosing
- Trial 2 for static had much higher mean DO than other trials, suggesting the need for further monitoring of conditions
- Oxygen is susceptible to noise from the environment so its significance was lower than the biomass.

Cost Breakdown

Cost Factor (5 days)	RL	PID	Static
Light Energy Cost	$0.122	$0.142	$0.146
Nutrient Cost	$0.012	$0.015	$0.016
Computing Power	$0.094	$0.075	$0.072
Total Operational	$0.228	$0.231	$0.234
Biomass per Dollar (g/$)	*10.95*	*7.75*	*5.53*

Discussion

Continual RL improves oxygen output and biomass by dynamically adapting light and nutrient conditions, reducing the inefficiencies seen in static control.

Why?

Avoids photoinhibition at sustained high intensities
- 63% avg compared to 75%

Adaptive timing injects nutrients when needed
- used 25% nutrients

Limitations

Wet mass proxy

Ambient Temperature

Lab-scale only (250ml)

Implications

Low-cost deployment

Carbon Capture

General framework

Biofuel Production

Future Work

Multi-strain Testing

Pilot-scale Deployment

Salt & Heat Control

Digital Twin Integration

References

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See binder for full list of sources

Physical Prototype

Physical prototype displayed at booth

